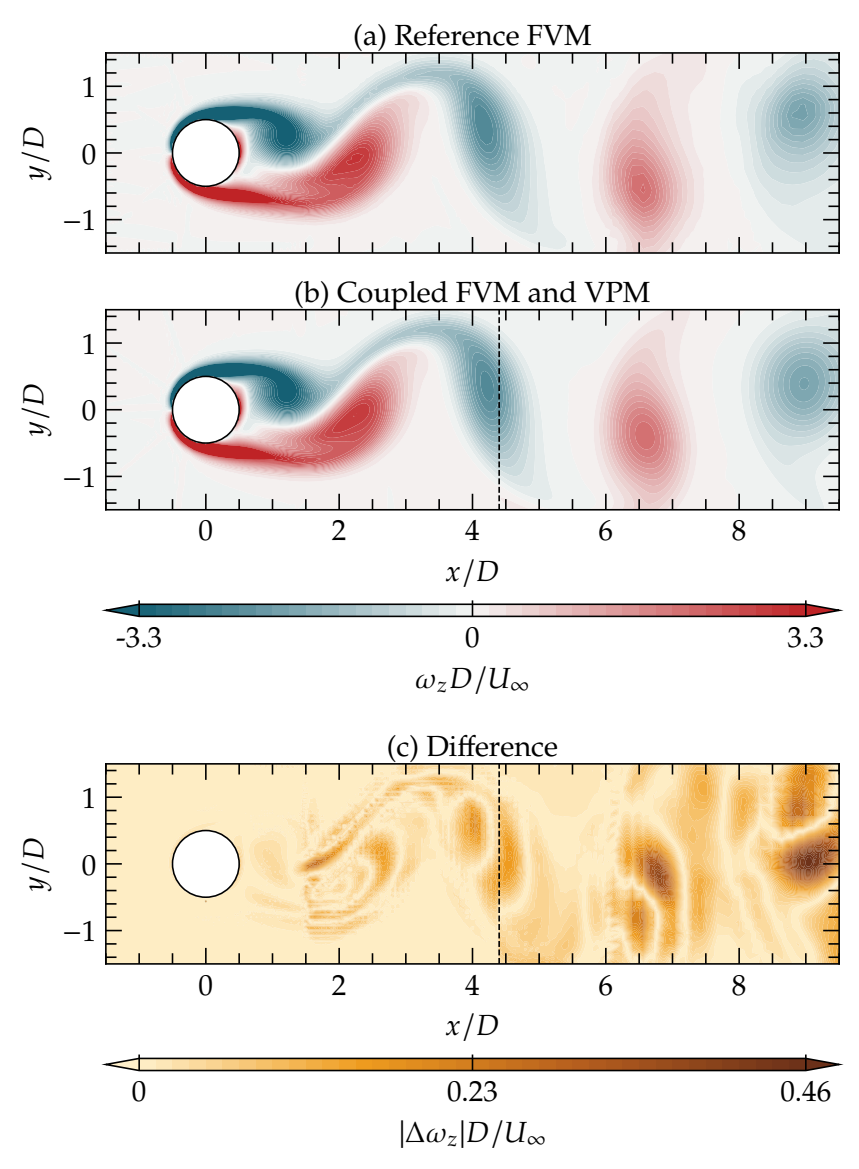

(a) Reference FVM
(b) Coupled FVM and VPM
(c) Difference
y/D
x/D
1
0
−1
0
2
4
6
8
-3.3
0
3.3
$\omega_z D/U_\infty$
0
0.23
0.46
$|\Delta\omega_z| D/U_\infty$